

Fig. 2: Power supply circuit

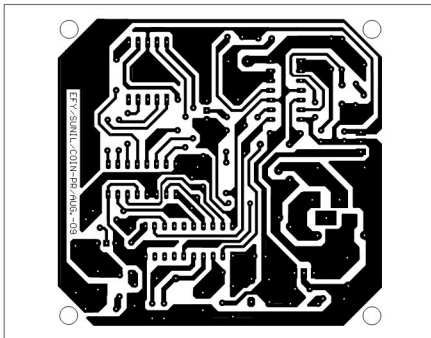


Fig. 3: Actual-size, single-side PCB for the put-coin-and-draw-power circuit

is precisely controlled by an external resistor and capacitor. The two timers operate independently of each other, sharing only V_{CC} and ground. The circuits may be triggered and reset on falling waveforms. One timer of NE555 is used for coin detection.

LDR1, connected at trigger pin 6 of IC1, offers low resistance when light is falling on it and its trigger input goes low to set the flip-flop and make output pin 5 of IC1 high.

When a coin is inserted, it interrupts the light falling on LDR1, and trigger pin 6 of IC1 goes high to make output pin 5 low. This high-to-low

pulse is used by the microcontroller to display the coin count.

Microcontroller AT89C2051 is the heart of the circuit. It is a low-voltage, high-performance, 8-bit microcontroller that features 2 kB of Flash, 128 bytes of RAM, 15 input/output (I/O) lines, two 16-bit timers/counters, a five-vector two-level interrupt architecture, a full-duplex serial port, a precision analogue comparator, on-chip oscillator and clock circuitry. A 12MHz crystal is used for providing the basic clock frequency. All I/O pins are reset to '1' as soon as RST goes high. Holding RST pin high for two

PARTS LIST

Semiconductors:

- IC1 - NE555 dual timer
- IC2 - AT89C2051 microcontroller
- IC3 - CD4511 7-segment decoder/driver
- IC4 - 7805 5V regulator
- IC5 - 7806 6V regulator
- T1, T2 - BC337 npn transistor
- D1-D5 - 1N4007 rectifier diode
- LED1-LED5 - 5mm LED
- DIS1 - LT543 common-cathode, 7-segment display

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1 - 220-ohm
- R2 - 33-kilo-ohm
- R3 - 220-kilo-ohm
- R4, R7, R9, R25 - 330-ohm
- R5, R8 - 1-kilo-ohm
- R6 - 10-kilo-ohm
- R10-R16 - 270-ohm
- R17-R24 - 4.7-kilo-ohm
- VR1 - 2.2-mega-ohm preset

Capacitors:

- C1, C7 - 10µF, 16V electrolytic
- C2, C3 - 0.01µF ceramic disk
- C4 - 100µF, 16V electrolytic
- C5, C6 - 33pF ceramic disk
- C8 - 1000µF, 35V electrolytic
- C9, C10 - 0.1µF ceramic disk

Miscellaneous:

- X1 - 230V AC primary to 9V, 500mA secondary transformer
- RL1 - 6V, 1C/O relay
- LDR1 - 10mm
- S1 - Push-to-on switch
- S2 - On/off switch
- Coin collection assembly

machine cycles, while the oscillator is running, resets the device. Power-on reset is derived from resistor R6 and capacitor C7. Switch S1 is used for manual reset.

Coin-detection output pin 5 of NE555 is interfaced with port pin P3.0 of the microcontroller (IC2). The microcontroller program counts the number of coins inserted and the count is shown on a 7-segment display.

The 'A' through 'D' inputs of 7-segment decoder IC3 are interfaced with port pins P1.4 through P1.7 of IC2. IC3 accepts the BCD input and decodes it to show on the 7-segment display. Coin detection is also indicated by LED2, which is connected to pin P3.7 of the microcontroller.